

Exam 2 Section Review— PSTAT 5LS

Spring 2026

Multiple Choice

Use the following information to answer Questions 1 to 4:

How difficult would it be for you to give up social media? In a random sample of 1316 U.S. teens aged 13 to 17, 54% of them reported that it would be hard for them to give up social media. We would like to construct a confidence interval for p , where p represents the proportion of all U.S. teens who would report that it would be hard for them to give up social media.

1. The observations are independent of one another because this was a random sample. The other condition to construct a confidence interval for the sample proportion p is satisfied because the:
 - ☐ sample size is greater than 30 with not much skew.
 - ☐ observed number of successes and failures are both at least 10.
 - ☐ expected number of successes and failures are both at least 10 when the null hypothesis is true.

2. Which of the following is the 98% confidence interval for the proportion of all U.S. teens who would report that it would be hard for them to give up social media?
 - ☐ 0.505 to 0.575
 - ☐ 0.508 to 0.572
 - ☐ 0.513 to 0.567
 - ☐ 0.517 to 0.563
 - ☐ This cannot be determined without more information.

3. Which of the following is true?
 - ☐ The 98% confidence interval provides a range of reasonable values for the sample proportion of U.S. teens who would report that it would be hard for them to give up social media.
 - ☐ The 98% confidence interval provides a range of reasonable values for the population proportion of U.S. teens who would report that it would be hard for them to give up social media.
 - ☐ The sample proportion of U.S. teens who would report that it would be hard for them to give up social media will be in the confidence interval 98% of the time.
 - ☐ The population proportion of U.S. teens who would report that it would be hard for them to give up social media will be in the confidence interval 98% of the time.

4. Suppose that you are asked to construct a 95% confidence interval using these data. How would the margin of error for the 95% confidence interval compare to the margin of error for the 98% confidence interval you constructed above? The margin of error for the 95% confidence interval would be:
 - ☐ larger than the margin of error for the 98% confidence interval.
 - ☐ smaller than the margin of error for the 98% confidence interval.
 - ☐ the same as the margin of error for the 98% confidence interval.

Use the following information to answer Questions 5 to 7:

“Animal Crossing: New Horizons” is a popular game for the Nintendo Switch. Researchers at Nintendo would like to know more about who is playing the game regularly and has some scenarios listed. Identify the parameter of interest.

5. Researchers want to estimate the number of hours spent playing Animal Crossing in a week, on average. The parameter is:
- ☐ p .
 - ☐ μ .
 - ☐ $\mu_1 - \mu_2$.
6. Researchers want to see if more than half of all Animal Crossing players have played the game for more than 500 hours over the life of the game. The parameter is:
- ☐ p .
 - ☐ μ .
 - ☐ $\mu_1 - \mu_2$.
7. Researchers wish to estimate the difference in the average amount of time spent playing Animal Crossing across Nintendo Switch users who are between the ages of 18-30 and Nintendo Switch users who are between the ages of 30-45. The parameter is:
- ☐ p .
 - ☐ μ .
 - ☐ $\mu_1 - \mu_2$.
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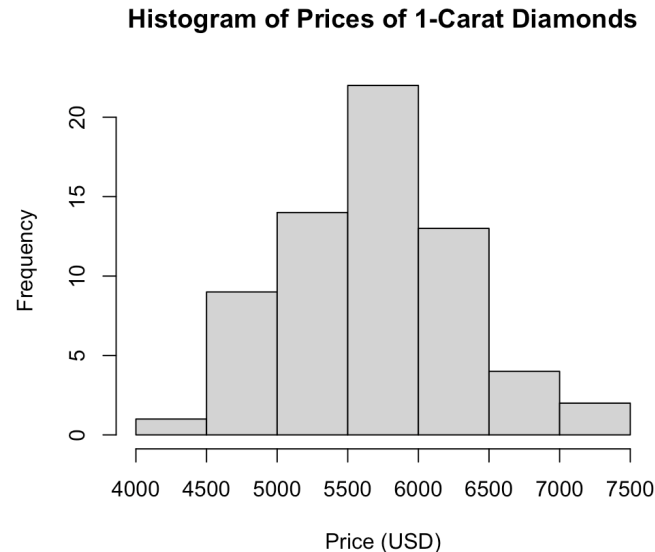
Use the following information to answer Questions 8 to 9:

Nathaniel, an intern at a national gym franchise, has been tasked with estimating the average number of hours that Gen Z'ers (those born between 1997 and 2012) spend exercising each week. Nathaniel obtained a random sample of 78 Gen Z'ers and found the sample mean to be 4.1 hours with a sample standard deviation of 2.1 hours.

8. Which of the following is the margin of error for a 90% confidence interval for the population mean number of hours that Gen Z'ers spend exercising each week?
- ☐ 0.04
 - ☐ 0.16
 - ☐ 0.39
 - ☐ 0.48
9. What would happen to the width of the confidence interval if Nathaniel increase the sample size but kept the confidence level the same?
- ☐ The width of the confidence interval would increase.
 - ☐ The width of the confidence interval would decrease.
 - ☐ The width of the confidence interval would stay the same.
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Free Response

10. **1-Carat Diamonds** An online jeweler has a large collection of diamonds. A random sample of 65 diamonds that were listed as 1-carat diamonds was taken from their website. The average price for these 65 diamonds was \$5671.30 with a standard deviation of \$648.80.



- One of the conditions for inference is that we have a random sample of diamonds. Check the remaining condition for inference.
- Which of the following do we need to use to find the critical value for a 99% confidence interval?
 - ☐ `pt(0.01, df = 64, lower.tail = FALSE)`
 - ☐ `pt(0.005, df = 64, lower.tail = FALSE)`
 - ☐ `qt(0.01, df = 64, lower.tail = FALSE)`
 - ☐ `qt(0.005, df = 64, lower.tail = FALSE)`
- Construct a 99% confidence interval for the population mean price for 1-carat diamonds offered by this online retailer. Make sure to show your work. For your reference, the critical value is 2.655. (Note: you do not need to interpret the confidence interval.).
- Refer back to your confidence interval. Would you suggest the jeweler have a mean price of \$5900 for 1-carat diamonds at their store? Clearly state “yes” or “no” and briefly (in 1-2 sentences) explain.

11. **Lemur Weights** The Duke Lemur Center in Durham, North Carolina, is "a world leader in the study, care, and protection of lemurs—Earth's most threatened group of mammals."¹

Lemurs are native to Madagascar, and the Duke Lemur Center has the largest (and most diverse) population of lemurs outside of Madagascar.

According to the Wisconsin National Primate Research, female ring-tailed lemurs in the wild have a mean weight of 2213 grams. Does the average weight for female ring-tailed lemurs in captivity differ from the mean of 2213 grams for female ring-tailed lemurs in the wild?

A random sample of 53 female ring-tailed lemurs at the Duke Lemur Center was taken, and the lemurs were weighed.² The mean was 2406.557 grams, and the standard deviation was 378.696 grams.

Here is a histogram of the distribution of weights for the sample of lemurs:



- State the null and alternative hypotheses for this hypothesis test. Be sure to define the parameter in context.
- Assume that the weights of the lemurs are independent of one another. Check the remaining condition for inference.
- Calculate the test statistic.
- Think about the value of the test statistic you just calculated. How do you think the p-value for this test will compare to $\alpha = 0.05$? Clearly state if you think the p-value will be "smaller," "larger", or "about the same" and briefly (in 1-2 sentences) support your decision.
- Based on the results of the hypothesis test and what you think about the size of the p-value, write a conclusion for this hypothesis test at the $\alpha = 0.05$ significance level. Be sure to include:
 - Your decision to reject or fail to reject the null hypothesis (and why)
 - Whether or not the results are statistically significant
 - A clear interpretation of the result in the context of the study
- If you made a mistake with your decision, what type of testing error would it be?
 - ☐ Type I Error
 - ☐ Type II Error

¹<https://lemur.duke.edu/>

²**Note:** We will assume that information about ring-tailed lemurs at the Duke Lemur Center can be used to generalize to the population of all ring-tailed lemurs in captivity.